CLASS 10 MATHEMATICS • CHAPTER 2

Polynomials

Exhaustive point-by-point presentation covering every concept, geometric meaning, and relationship between zeroes and coefficients directly from the textbook.

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Revisiting Polynomials and Their Degrees

Degree Definition

- If p(x) is a polynomial in x, the **highest power of x** in p(x) is called its degree.
- Example: $4x + 2$ is of degree 1.
- $2y^2 - 3y + 4$ is of degree 2.
- $5x^3 - 4x^2 + x - \sqrt{2}$ is of degree 3.

What is NOT a Polynomial?

- Expressions with negative or fractional exponents in the denominator/variable are not polynomials.
- Examples: $1 / (x - 1)$, $\sqrt{x} + 2$, and $1 / (x^2 + 2x + 3)$ are **not polynomials**.

Section 2.1 Live Check

Q1. What is meant by the degree of a polynomial? Give an example of a polynomial of degree 4.

Q2. Why is an expression like $1 / (x - 1)$ not considered a polynomial?

Q3. Can you state the highest power in $7u^6 - (3/2)u^4 + 4u^2 + u - 8$? What is its degree?

Linear, Quadratic, and Cubic Polynomials

Linear & Quadratic

- **Linear Polynomial (Degree 1):** General form $ax + b$ ($a \neq 0$). E.g., $2x - 3$, $\sqrt{3}x + 5$.
- **Quadratic Polynomial (Degree 2):** Derived from 'quadrate' (square). General form $ax^2 + bx + c$ ($a \neq 0$).

Cubic Polynomial

- **Cubic Polynomial (Degree 3):** General form $ax^3 + bx^2 + cx + d$ ($a \neq 0$).
- Examples: $2 - x^3$, x^3, $\sqrt{2}x^3$, $3x^3 - 2x^2 + x - 1$.

Classification Live Check

Q1. What condition must be satisfied by coefficient 'a' in a quadratic polynomial $ax^2 + bx + c$?

Q2. Why is a polynomial of degree 2 called 'quadratic'? Where does the word come from?

Q3. Give the general standard form of a cubic polynomial.

Value of a Polynomial at x = k and Zeroes

Definition of a Zero

A real number k is said to be a **zero of a polynomial p(x)** if p(k) = 0.

Example: p(x) = x² - 3x - 4

- p(2) = 2² - 3(2) - 4 = -6 (value at x = 2).
- p(-1) = (-1)² - 3(-1) - 4 = **0**.
- p(4) = 4² - 3(4) - 4 = **0**.
- Thus, -1 and 4 are zeroes of x² - 3x - 4.

Zero of a Linear Polynomial

- For p(x) = ax + b, setting ak + b = 0 gives k = -b / a.
- Zero = -(Constant term) / (Coefficient of x).

Zeroes Concept Live Check

Q1. What does it mean when we say that $p(k) = 0$ for a real number k?

Q2. How do you find the zero of a linear polynomial $ax + b$?

Q3. Verify whether $x = 3$ is a zero of the polynomial $p(x) = x^2 - 9$.

Geometrical Meaning of Zeroes (Linear & Quadratic)

Linear Polynomial

- The graph of $y = ax + b$ is a straight line.
- It intersects the x-axis at exactly one point, namely $(-b / a, 0)$.
- The x-coordinate of this intersection point is the unique zero.

Quadratic Polynomial

- The graph of $y = ax^2 + bx + c$ is a **parabola**.
- Opens upwards if $a > 0$, downwards if $a < 0$.
- Zeroes are the x-coordinates of the points where the parabola intersects the x-axis.

Geometrical Meaning Live Check

Q1. What shape does the graph of a quadratic polynomial $ax^2 + bx + c$ take?

Q2. How do you determine the zeroes of a polynomial purely by looking at its graph?

Q3. When does a parabola open upwards versus downwards? What controls this?

Possible Intersections and Cubic Polynomials

Three Cases for Quadratics

- **Case (i):** Intersects x-axis at two distinct points (2 zeroes).
- **Case (ii):** Touches x-axis at one point (1 equal/coincident zero).
- **Case (iii):** Completely above or below x-axis (no zero).

Cubic Polynomials & General Rule

- Cubic curves intersect the x-axis at at most 3 points (at most 3 zeroes).
- **General Rule:** A polynomial p(x) of degree n has **at most n zeroes**.

Graph Cases Live Check

Q1. What is the maximum number of zeroes a polynomial of degree n can have?

Q2. Explain what happens graphically when a quadratic polynomial has no real zeroes.

Q3. Can a cubic polynomial have only one real zero? Give a graphical reason.

Relationship for Quadratic Polynomials

Quadratic Zeroes & Coefficients Formula

If α and β are the zeroes of $ax^2 + bx + c$ $(a \neq 0)$, then:

Sum of zeroes $(\alpha + \beta) = -b / a = -(\text{Coefficient of } x) / (\text{Coefficient of } x^2)$

Product of zeroes $(\alpha\beta) = c / a = (\text{Constant term}) / (\text{Coefficient of } x^2)$

Quadratic Relations Live Check

Q1. What is the sum of zeroes for a quadratic polynomial $ax^2 + bx + c$?

Q2. What is the product of zeroes expressed in terms of coefficients?

Q3. If a quadratic polynomial has zeroes α and β, what are the factors of the polynomial?

Relationship for Cubic Polynomials

Cubic Zeroes & Coefficients

If α, β, γ are zeroes of $ax^3 + bx^2 + cx + d$ $(a \neq 0)$, then:

$$\alpha + \beta + \gamma = -b / a = -(\text{Coefficient of } x^2) / (\text{Coefficient of } x^3)$$
$$\alpha\beta + \beta\gamma + \gamma\alpha = c / a = (\text{Coefficient of } x) / (\text{Coefficient of } x^3)$$
$$\alpha\beta\gamma = -d / a = -(\text{Constant term}) / (\text{Coefficient of } x^3)$$

Cubic Relations Live Check

Q1. How many zeroes can a cubic polynomial have at most?

Q2. What is the formula for the sum of products of zeroes taken two at a time for a cubic polynomial?

Q3. What is the product of all three zeroes of a cubic polynomial $ax^3 + bx^2 + cx + d$?

Worked Exercise Highlights

Problem 1: Zeroes & Verification

- Find zeroes of $x^2 - 2x - 8$.
- Split middle term: $(x - 4)(x + 2) = 0$.
- Zeroes: $x = 4, x = -2$.
- Sum $= 2 = -(-2)/1$; Product $= -8 = -8/1$. **Verified!**

Problem 2: Quadratic from Sum/Product

- Given sum $= 1/4$, product $= -1$.
- Let polynomial be $k[x^2 - (\text{sum})x + \text{product}]$.
- $k[x^2 - (1/4)x - 1] = k/4\,(4x^2 - x - 4)$.
- For $k = 4$, polynomial is $\mathbf{4x^2 - x - 4}$.

Exercise Highlights Live Check

Q1. How do we split the middle term for factoring a quadratic polynomial?

Q2. Given the sum and product of zeroes, how can you immediately construct the quadratic polynomial?

Q3. Why is there a constant multiplier 'k' when constructing a polynomial from its zeroes?

Chapter Recap & Checklist

Degrees & Graphs

- **Polynomial Degrees:** Linear (1), Quadratic (2), Cubic (3).
- **Geometrical Meaning:** Zeroes are x-coordinates of intersection with the x-axis.
- **Max Zeroes:** Degree n polynomial has at most n zeroes.

Coefficient Relations

- **Quadratic:** $\alpha + \beta = -b/a$, $\alpha\beta = c/a$.
- **Cubic:** $\alpha + \beta + \gamma = -b/a$, $\alpha\beta + \beta\gamma + \gamma\alpha = c/a$, $\alpha\beta\gamma = -d/a$.

THANK YOU

"Algebra is generous; she often gives more than is asked of her." — Jean le Rond d'Alembert

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